

# BDRTK-980 GNSS Board

## Product Manual



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# General Notes

Name: RTK GNSS Board

Model: BDRTK-980

## User Manual

Thank you for choosing our company's BDRTK-980 GNSS Board series products. Before using this manual, please confirm that the model number corresponds to your hardware device. Prior to installing and using the BDRTK-980 GNSS Board (including both hardware and software), please read this manual and all relevant technical documentation for the system carefully.

In this manual, the "BDRTK-980 GNSS Board" is abbreviated as BDRTK-980. Depending on your specific application requirements, you may need to consult the following technical documents during use:

《UM980\_User Manual》

《UPrecise\_User Manual》

If you have any questions during use, please contact our company's technical support personnel.

Our company's Quality Management System has been audited and conforms to the GB/T 19001-2016 / ISO 9001:2015 standards.

Our company reserves all patent rights and the final interpretation rights for all materials contained in this manual.

## 1. Product Introduction

The BDRTK-980 is a high-precision GNSS board independently developed and manufactured by our company. Built around the advanced Unicore UM980 module, it supports GPS, BDS, GLONASS, Galileo, QZSS, NavIC, SBAS and L-Band, and leverages multi-frequency anti-jamming technology to deliver fast initialization, high accuracy, and reliable RTK performance in challenging environments. Designed for ease of integration, the board features direct USB, RS232, and TTL interfaces, making it an ideal and convenient solution for efficiently setting up base or rover stations across a wide range of applications, including surveying and mapping, autonomous robots and lawnmowers, UAVs, precision agriculture, and intelligent driving.

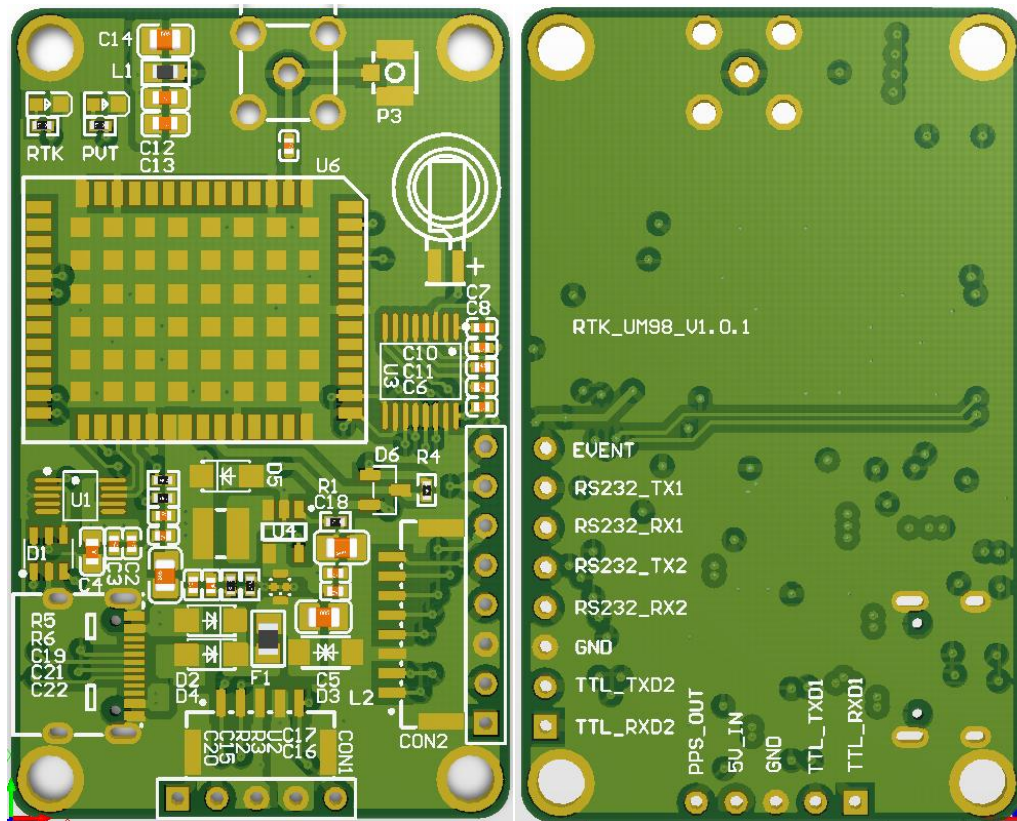
## 2. Key Features

- Achieves centimeter-level RTK positioning accuracy up to 0.8 cm.
- Utilizes the Unicore-proprietary NebulasIV SoC that integrates RF, baseband, and a high-precision algorithm.
- All-constellation multi-frequency RTK engine and advanced RTK technology.
- Features instant RTK initialization technology for significantly reduced convergence times.
- Incorporates advanced anti-jamming technology with 60 dB suppression and detection capabilities.
- Provides reliable heading information through integrated Heading2 orientation technology.
- Delivers high-precision positioning in standalone mode without relying on a base station.
- Supports multiple PPP services including B2b-PPP, E6-HAS, and QZSS L6E (MADOCa).
- Offers plug-and-play connectivity via direct USB, RS232, and TTL interfaces.

### 3. Technical Specifications

| No. | Parameter Name          | Technical Specifications                                                                                                                                           |
|-----|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Channel                 | 1408 channels, based on NebulasIV                                                                                                                                  |
| 2   | Frequency               | GPS L1C/A, L1C, L2C, L2P(Y), L5; BDS B1I, B2I, B3I, B1C, B2a, B2b; GLONASS G1, G2, G3; Galileo E1, E5a, E5b, E6 QZSS L1C/A, L1C, L2C, L5, L6; NavIC L5; SBAS L1C/A |
| 3   | Cold Start TTFF         | < 12 s                                                                                                                                                             |
| 4   | Hot Start TTFF          | < 4 s                                                                                                                                                              |
| 5   | positioning accuracy    | Single Point Positioning(RMS):<br>Horizontal: 1.5 m<br>Vertical: 2.5 m                                                                                             |
|     |                         | Differential positioning(RMS):<br>Horizontal: 0.4 m<br>Vertical: 0.8 m                                                                                             |
|     |                         | RTK Positioning(RMS)<br>Horizontal: 0.8 cm + 1 ppm<br>Vertical: 1.5 cm + 1 ppm                                                                                     |
| 6   | Time Accuracy (RMS)     | 20 ns                                                                                                                                                              |
| 7   | Velocity Accuracy (RMS) | 0.03 m/s                                                                                                                                                           |
| 8   | Data Update Rate        | < 50 Hz                                                                                                                                                            |
| 9   | Data Format             | NMEA 0183                                                                                                                                                          |
| 10  | Baud Rate               | 115200bps                                                                                                                                                          |
| 11  | Operating Temperature   | -40 ℃ ~ 85 ℃                                                                                                                                                       |
| 12  | Storage Temperature     | -55℃ ~ 95℃                                                                                                                                                         |
| 13  | Differential Data       | RTCM V3.X                                                                                                                                                          |
| 14  | Voltage                 | DC 4.0-5.5V    Typical: 5.0V                                                                                                                                       |
| 15  | Current                 | 160mA/5.0V                                                                                                                                                         |
| 16  | Dimension               | 32 x 52 x 11 mm                                                                                                                                                    |
| 17  | Weight                  | 11.0 g                                                                                                                                                             |
| 18  | Connector               | 1* 1.25 pitch 8pin with lock seat<br>1* 1.25 pitch 5pin with lock seat<br>1* Type-C USB Port<br>1* SMA interface                                                   |

#### 4. Pin Definition



| Pin Name  | Full Name & Meaning      | Functional Description                                                                                                    |
|-----------|--------------------------|---------------------------------------------------------------------------------------------------------------------------|
| RS232_TX1 | RS232 Transmit Channel 1 | Data output pin compliant with the RS232 standard. Transmits data from the module to external devices (e.g., a computer). |
| RS232_RX1 | RS232 Receive Channel 1  | Data input pin compliant with the RS232 standard.<br>Receives commands or data from external devices (e.g., a computer).  |
| RS232_TX2 | RS232 Transmit Channel 2 | Second independent RS232 data output channel,<br>with the same function as TX1                                            |
| RS232_RX2 | Receive Channel 2        | Second independent RS232 data input channel,<br>used for receiving configuration commands or other data.                  |

|          |                         |                                                                                                                                                                     |
|----------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GND      | Ground                  | Common reference ground for the circuit, providing the baseline potential for all signals.The return path for current for all interfaces.                           |
| TTL_TXD2 | TTL Transmit Channel 2  | TTL-level data output pin. Transmits data from the module to external devices.                                                                                      |
| TTL_RXD2 | TTL Receive Channel 2   | TTL-level data input pin.<br>Receives commands or data from external devices.                                                                                       |
| EVENT    | Event Marker            | Pin Outputs a pulse signal when a specific event occurs (e.g., reaching a precise timestamp). Often used for precision timing and synchronization applications.     |
| PPS_OUT  | Pulse Per Second Output | Outputs a precise pulse signal every second for time synchronization.The rising edge of the pulse is typically aligned precisely with the start of each UTC second. |
| 5V_IN    | 5 Volts Power Input     | Used to supply 5V DC power to the module from an external source as same as VCC.                                                                                    |
| TTL_TXD1 | TTL Transmit Channel 1  | First TTL-level data output channel (typically the main output channel).                                                                                            |
| TTL_RXD1 | TTL Receive Channel 1   | First TTL-level data input channel (typically the main configuration channel).                                                                                      |